6.4.2 Fundamental particles

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	particles and antiparticles; electron–positron, proton-antiproton, neutron-antineutron and neutrino-antineutrino	HSW7, 9
(b)	particle and its corresponding antiparticle have same mass; electron and positron have opposite charge; proton and antiproton have opposite charge	
(c)	classification of hadrons; proton and neutron as examples of hadrons; all hadrons are subject to both the strong nuclear force and the weak nuclear force	
(d)	classification of leptons; electron and neutrino as examples of leptons; all leptons are subject to the weak nuclear force but not the strong nuclear force	HSW7, 9

(e) simple quark model of hadrons in terms of up (u), down (d) and strange (s) quarks and their respective anti-quarks

(f) quark model of the proton (uud) and the neutron (udd)

(g) charges of the up (u), down (d), strange (s), anti-up ($\overline{u}$), anti-down ($\overline{d}$) and the anti-strange ($\overline{s}$) quarks as fractions of the elementary charge *e*

(h) beta-minus (β^-) decay; beta-plus (β^+) decay

(i) β^- decay in terms of a quark model;

$d \rightarrow u + {}^{0}_{-1}e + \overline{\nu}$

(j) β^+ decay in terms of a quark model;

$u \rightarrow d + {}^{0}_{+1}e + \nu$

(k) balancing of quark transformation equations in terms of charge

(l) decay of particles in terms of the quark model.